Tate's Thesis

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Contents

1 Local Functional Equation

To treat the local field of number fields, we fix the notation for sake of convience.

Definition 1.1. Let F be a field, then we call F is a local field if it is one of the following types:

1. Archimedean local field: $\mathbb{R}$ or $\mathbb{C}$;
2. Non-archimedean local field: a finite extension of $\mathbb{Q}_p$

Equivalently, a local field is a completion of a number field at a place, and it is non-archimedean if the place is finite.

In the case of non-archimedean local field, we use the following notation:

$$\begin{aligned} \mathcal{O} &: \text{ the valuation ring} \\ \mathcal{O}^\times &: \text{ the unit group of } \mathcal{O} \\ \mathfrak{p} &: \text{ the maximal ideal of } \mathcal{O} \\ \varpi &: \text{ a uniformizer} \\ \mathfrak{d} &: \text{ the different of } F/\mathbb{Q}_p \end{aligned}$$

Definition 1.2. Let F be a local field, then $f : F \to \mathbb{C}$ is called a Schwartz-Bruhat function if it is one of the following cases:

- When F is archimedean, f is a Schwartz function, which is a smooth and rapidly deacrasing function of class C^∞.
- When F is non-archimedean, f is locally constant and compactly supported.

Let $\mathcal{S}(F)$ denotes the $\mathbb{C}$-vector space of all Schwartz-Bruhat functions on F.

Definition 1.3. A LCA group is a locally compact hausdorff abelian group.

1.1 Additive Character

Let F be a local field, then in each case we can know that $(F, +)$ is a locally compact abelian group, which induces a Haar measure μ on F, then for any non-zero element $a \in F$, we can define a new measure on F by pull back:

$$\forall E \subset F \text{ measurable }, \quad \mu_a(E) = \mu(aE)$$

and we can conculde the following result:

Lemma 1.1. μ_a is a Haar measure on $(F, +)$, and there exists a constant factor $|a|$ such that $\mu_a = |a|\mu$, such a factor is independent of the choice of μ with

$$|a| = \begin{cases} \text{the standard absolute value if } F = \mathbb{R} \\ \text{the square of standard absolute value if } F = \mathbb{C} \\ N(\mathfrak{p})^{-v_{\mathfrak{p}}(a)} \text{ if } F \text{ is non-archimedean} \end{cases}$$

Proof. The case of $F = \mathbb{R}$ is clear. If $F = \mathbb{C}$, we take $a = u + iv$ and μ be the lebesgue measure, let $z = x+iy$ and then $az = (ux-vy)+i(vx+uy)$, and we consider the integration in $\mathbb{R}^2$, then the jacobian of the transformation is

$$|\det J_a| = \begin{vmatrix} u & -v \\ v & u \end{vmatrix} = u^2 + v^2 = \|a\|^2$$

so that shows $d(az) = \|a\|^2 dz$. If F is non-archimedean, we take μ be the Haar measure such that $\mu(\mathcal{O}) = 1$, and we set $q = N(\mathfrak{p}) = \#(\mathcal{O}/\mathfrak{p})$, then

$$1 = \mu(\mathcal{O}) = \mu(\bigsqcup_{i=1}^{q} x_i + \mathfrak{p}) = \sum_{i=1}^{q} \mu(x_i + \mathfrak{p}) = q\mu(\mathfrak{p})$$

so it shows that $\mu(\mathfrak{p}) = 1/q$, and same method shows that $\mu(\mathfrak{p}^n) = 1/q^n$. So for any $a \in F^\times$, we can write $a = \varpi^n u$ with $n = v_\mathfrak{p}(a)$ and $u \in \mathcal{O}^\times$, then $a\mathcal{O} = \varpi^n\mathcal{O} = \mathfrak{p}^n$ and then

$$\mu(a\mathcal{O})/\mu(\mathcal{O}) = 1/q^n = N(\mathfrak{p})^{-v_\mathfrak{p}(a)}$$

□

The constant factor can be extended to the whole field F by setting $|0| = 0$, and we call it **(normalized) module** on F. A obeservation here is that module is a power of standard absolute value on F, so $a \mapsto |a|$ defines a a continous map from F to $[0, \infty)$.

Remark. More generally, for any LCA group $(G, +)$ with Haar measure μ, if φ is a continuous automorphism of G, then $\mu \circ \varphi$ is also a Haar measure with $\mu \circ \varphi = \mathrm{mod}_G(\varphi) \cdot \mu$ for some constant factor only depending on φ, such a factor is called the module of φ on G. Let $\mathrm{Aut}(G)$ denote the group of all continuous automorphisms of G, then $\varphi \mapsto \mathrm{mod}_G(\varphi)$ is actually a homomorphism from $\mathrm{Aut}(G)$ to $\mathbb{R}_{>0}$. In the case of local field F, $F^\times$ can be embedded into $\mathrm{Aut}(F)$ by $a \mapsto m_a$ with $m_a(x) = ax$, and the module $\mathrm{mod}_F(m_a)$ is exactly $|a|$ defined in above lemma.

Definition 1.4. Additive character of a local field F is a continous group homomorphism $\psi : (F, +) \to \mathbb{S}^1$, and $\hat{F}$ is the set of all additive characters of F, it is called the dual group of F.

The dual group is indeed a group under pointwise multiplication, for topology on $\hat{F}$, we consider the compact-open topology, i.e. the topology generated by

$$[K, U] = \{\chi \in \hat{F} : \chi(K) \subset U\}$$

for every compact subset K of F and every open subset U of $\mathbb{S}^1$. It is a natural choice for functiona space, so dual group is ineed a topological group, a remarkable result here is that local field is self-dual.

Theorem 1.2. Let ψ be a non-trival additive character of F, then the map

$$\Psi : F \to \hat{F}, \quad a \mapsto \psi_a : x \mapsto \psi(ax)$$

is an isomorphism of LCA groups.

Proof. The proof depends on a fact that all local fields are a metrizable space such that compact subset is bounded. It is not hard to check that Ψ is a injective homomorphism, so firstly we prove that Ψ is continous: Let $[K, U]$ be an open basis of $\hat{F}$, then

$$\Psi^{-1}([K, U]) = \{a \in F : \psi_a(K) \subset U\} = \{a \in F : aK \subset \psi^{-1}(U)\}$$

notice that $\psi^{-1}(U)$ is a open since ψ is continous, so by continous map $m(x, y) = xy$ on $F \times F$, for any $y \in K$ there exists an neiborhood A_y of y and a neiborhood B_y of a such that $B_y A_y \subset \psi^{-1}(U)$. Let y runs over all K, and by compactness of K, there exists a finite index set I such that $K \subset \cup_{i \in I} A_i$, and we take $V_a = \cap_{i \in I} B_i$ as a neiborhood of a such that $V_a \subset \Psi^{-1}[K, U]$, it shows that a is an interior and then $\Psi^{-1}[K, U]$ is open.

Secondly, we prove that Ψ is an homemorphism to its image, it is enough to prove the image of open ball B_ϵ centred at 0 with radius ϵ is open in $\hat{F}$. Non-trival character ψ implies an element $b \in F$ such that $\psi(b) \neq 1$, and we construct a neiborhood of trivial character by $[\overline{B_R(0)}, W] \cap \Psi(F)$, where W is an neiborhood of 0 not cotaining $\psi(b)$, and $R \geq \|b\|/\epsilon$, then this neiborhood is contained in $\Psi(B_\epsilon)$. For other non-trival character in image, a translation of group allows us to conclude the result.

Finally, $\Psi(F)$ is complete since F is complete, and it is closed since $\hat{F}$ is hausdorff, so it is enough to prove that $\Psi(F)$ is dense in $\hat{F}$, that will imply $\Psi(F) = \hat{F}$. In fact, $x \in \Psi(F)^\perp$ implies that $\Psi(ax) = 1$ for any $a \in F$, and it shows that the annihilator is zero, so $\Psi(F)$ is dense in $\hat{F}$. □

Remark. A LCA group is not necessary self-dual, for example the dual group of $\mathbb{Z}$ is $\mathbb{S}^1$, and the dual group of $\mathbb{S}^1$ is $\mathbb{Z}$.

By the theory of harmonic analysis on LCA, if LCA group G is self-dual, then it is nice to identify the fourier tranform of a function f on G as a function on G itself, so it motivates us to define the transform on local field as follows:

Definition 1.5. Fix a non-trival additive character ψ and a Haar measure μ on F, the Fourier transform of type (ψ, μ) of a function $f \in L^1(F)$ is defined as

$$\hat{f}(y) = \int_F f(x)\psi(xy)\, d\mu(x)$$

There is no confusion to identify y as ψ_y by isomorphism, and the norm of ψ is always 1, so $|\hat{f}(y)| \leq \|f\|_{L^1} < \infty$ ensures that the transform is well-defined.

Definition 1.6. The standard additive character of F is defined as following:

- If $F = \mathbb{R}$, let $\psi(x) = e^{-2\pi i x}$;
- If $F = \mathbb{Q}_p$, let $\psi(x) = e^{2\pi i \{x\}}$, where $\{x\}$ is the fractional part of x defined by

$$x = \sum_{n \geq N} a_n p^n \implies \{x\} = \sum_{n=N}^{-1} a_n p^n$$

  where $\{x\} = 0$ if $N \geq 0$, i.e. $x \in \mathbb{Z}_p$.

- Let $F_0 = \mathbb{R}$ or $\mathbb{Q}_p$, and F be a finite extension of F_0, let the standard additive character of F be the compistion
$$F \xrightarrow{\mathrm{Tr}_{F/F_0}} F_0 \xrightarrow{\psi_0} \mathbb{S}^1$$
where ψ_0 is the standard additive character of F_0.

In particular, if $F = \mathbb{Q}$, the standard character is $\psi(z) = e^{-2\pi i(z+\bar{z})}$. In the case of $F = \mathbb{Q}_p$, we have $\{x\} \in \mathbb{Z}[1/p]$ and $x - \{x\} \in \mathbb{Z}_p$, the intersection $\mathbb{Z}_p \cap \mathbb{Z}[1/p] = \mathbb{Z}$ implies that standard character ψ is a non-trival homomoprhism. In the aspect of continuity, we notice that $\mathbb{Z}_p \subset \ker \psi$ as an open subset containing 0, which shows that ψ is continous at 0, then ψ is continous on $\mathbb{Q}_p$ since it is a topological group. Hence ψ is exactly a non-trival additive character of $\mathbb{Q}_p$, It is not clear than the case $F = \mathbb{R}$.

From now on, we will always use standard character in the Fourier transform, so the transform will depend only on the choice of Haar measure, so a normalization can be done to obtain the inversion formula.

Proposition 1.3. Let F be a local field, then there exists a unique self-dual Haar measure μ, that means if $f \in \mathcal{S}(F)$, then $\hat{f} \in \mathcal{S}(F)$ and $\widehat{\hat{f}}(x) = f(-x)$, explicitly the measure is given as following:

$$dx = \begin{cases} \text{lebesgue measure of } \mathbb{R} & \text{If } F = \mathbb{R} \\ 2 \times \text{lebesgue measure of } \mathbb{C} & \text{If } F = \mathbb{C} \\ \text{the Haar measure such that } \mu(\mathcal{O}) = N(\mathfrak{d})^{-\frac{1}{2}} & \text{If } F \text{ is non-archimedean} \end{cases}$$

Proof. For any Haar measure μ, $\widehat{\hat{f}}(x) = cf(-x)$ holds for some constant c, so it is enough to choose a test function to modify. If $F = \mathbb{R}$, we choose $f(x) = e^{-\pi x^2}$ and calculate the transform under the lebesgue measure, than $\hat{f} = f$; If $F = \mathbb{C}$ we choose $f(x)$, and calculate the transform under $c \cdot dl$ (c times the lebesgue measure), than $\hat{f} = \frac{c}{2} f$ shows $2\,dl$ is the self-dual measure; If F is non-archimdean we choose $f = \mathbf{1}_{\mathcal{O}}$, then under some haar measure μ on a $\widehat{\widehat{\mathbf{1}_{\mathcal{O}}}} = \mu(\mathcal{O})\mu(\mathcal{O}^{\perp})\mathbf{1}_{\mathcal{O}}$, and we assume difference $\mathfrak{d} = \mathfrak{p}^m$ to calculate

$$\mu(\mathcal{O}^{\perp}) = \mu(\mathfrak{d}^{-1}) = \mu(\varpi^{-m}\mathcal{O}) = |\varpi|^{-m} N(\mathcal{O}) = N(\mathfrak{d})\mu(\mathcal{O})$$

so normization $\mu(\mathcal{O})\mu(\mathcal{O}^{\perp}) = 1$ gives the Haar measure. □

1.2 Multiplicative Character

The module on local field F induced a continous homomorphism

$$F^{\times} \to \mathbb{R}_{>0}, \quad x \mapsto |x|$$

the kernel of this homomorphism will be important in the following discussion, and we denote it by U, and we can conclude

$$U = \begin{cases} \{\pm 1\} & \text{If } F = \mathbb{R} \\ \mathbb{S}^1 & \text{If } F = \mathbb{C} \\ \mathcal{O}^{\times} & \text{If } F \text{ is non-archimedean} \end{cases}$$

The group U as a subspace of local field F is compact in each case, and also open if the F is non-archimedean.

Definition 1.7. a (multiplicative) quasi-character is a continous group homomorphism $\chi : F^\times \to \mathbb{C}^\times$, and the set of all quasi-characcters is denoted by $\mathcal{X}(F^\times)$.

a quasi-character χ is called a (unitary) character if $\mathrm{im}(\chi) \subset \mathbb{S}^1$; a quasi-character χ is called unramified if χ is trival on U.

The vocabulary here follows Tate's thesis, here are the examples in the case of archimedean local field, which is the classical result in harmonic analysis.

Example 1.4. Every quasi-character of $\mathbb{R}^\times$ is of the form

$$\chi_{\epsilon,s}(t) = \mathrm{sign}(t)^\epsilon \cdot |t|^s, \quad \epsilon \in \{0,1\}, s \in \mathbb{C}$$

Every quasi-character of $\mathbb{C}^\times$ is of the form

$$\chi_{n,s}(z) = (\frac{z}{\|z\|})^n \cdot |z|^s, \quad n \in \mathbb{Z}, s \in \mathbb{C}$$

where $\|z\|$ is the standard absolute value on $\mathbb{C}$, and in both case, the character is unitary if and only if s is a purely imaginary number.

By the example, it is natural to separate the quasi-character into two parts, the first part is a unitary character on units, and the second part is a power of module, in fact it is unramified and we can generalize the result.

Lemma 1.5. A quasi-character $\chi \in \mathcal{X}(F^\times)$ is unramified if and only if it is of the form $\chi(x) = |x|^s$ for some $s \in \mathbb{C}$. Let $\mathcal{U}(F^\times)$ denote the set of all unramified quasi-characters on $F^\times$, then it forms a subgroup of $\mathcal{X}(F^\times)$ with isomorphism

$$\mathcal{U}(F^\times) \cong \begin{cases} \mathbb{C} & \text{If } F \text{ is archimedean} \\ \mathbb{C}/\frac{2\pi i}{\log N(\mathfrak{p})}\mathbb{Z} & \text{If } F \text{ is non-archimedean} \end{cases}$$

Proof. Notice that χ is unramified if and only if χ factors through $|F^\times|$, because $U \subset \ker \chi$ if χ is trivial on U, while U is the kernel of the surjection $x \mapsto |x|$, that means there exists a continous homomorphism $\tilde{\chi} : |F^\times| \to \mathbb{C}^\times$ such that the following diagram commutes:

$$\begin{tikzcd} F^\times \arrow[r, "\chi"] \arrow[d, "x \mapsto |x|"'] & \mathbb{C}^\times \\ |F^\times| \arrow[ur, "\tilde{\chi}"'] & \end{tikzcd}$$

Hence if χ is unramified, it is enough to prove that continous homomorphism $\tilde{\chi} : |F^\times| \to \mathbb{C}^\times$ is of the form $x \mapsto x^s$ for some $s \in \mathbb{C}$. If F is archimedean, it is immediate by the example above, if F is non-archimedean, $|F^\times| = q^{\mathbb{Z}}$ with $q = N(\mathfrak{p})$, so the morphism is determined by the image of q, if we set $\tilde{\chi}(q) = a$, then we can choose $s = \log a$ with respect to a branch of logarithm.

Conversely, it is not hard to check that $\chi_s(x) = |x|^s$ is unramified for any $s \in \mathbb{C}$, which proves the first part of the lemma and induces a surjective map

$$\Phi : \mathbb{C} \to \mathcal{U}(F^\times), \quad s \mapsto \chi_s$$

and it is not hard to check $\mathcal{U}(F^\times)$ is indeed a subgroup of $\mathcal{X}(F^\times)$ and $\chi_{s+z} = \chi_s \cdot \chi_z$ for any $s, z \in \mathbb{C}$, which implies that Φ is a group homomorphism with kernel $\{s \in \mathbb{C} : |x|^s = 1, \forall x \in F^\times\}$. If F is archimedean, the kernel is trivial so Φ is an isomorphism; if F is non-archimedean, the kernel is exactly $\{s \in \mathbb{C} : |\varpi|^s = 1\} = \frac{2\pi i}{\log q}\mathbb{Z}$, so it induces a desired isomorphism. $\square$

Lemma 1.6. The map $\chi \mapsto (\chi|_U, \chi \cdot (\chi|_U \circ \rho)^{-1})$ gives an isomorphism

$$\mathcal{X}(F^\times) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{cont}}(U, \mathbb{S}^1) \times \mathcal{U}(F^\times)$$

where $\rho : F^\times \to U$ is a retraction depending on the local field F, and every quasi-character χ is of the form $\chi = \tilde{\chi}|\cdot|^s$ for some $s \in \mathbb{C}$ and a unitary character $\tilde{\chi}$ conciding with χ on U.

Proof. It is enough to prove the non-arcimedean case, since we have seen that $\rho(x) = x/\|x\|$ when F is archimedean. Notice that $U = \mathcal{O}^\times$ is a compact subgroup of $F^\times$, then the restriction $\chi|_U$ must be a unitary, and we define $\rho(x) = \varpi^{-v_{\mathfrak{p}}(x)}x$, which is a continous homomorphism with $\rho(\varpi^n u) = u$ for any $n \in \mathbb{Z}$ and $u \in \mathcal{O}^\times$. Hence $c(x) = \chi \cdot (\chi|_U \circ \rho)^{-1}$ defines a quasi-character with $c(u) = 1$ for any $u \in \mathcal{O}^\times$, by lemma 1.5 it shows that c is unramified of the form $c(x) = |x|^s$ for some $s \in \mathbb{C}$, so $\chi = \chi_u| \cdot |^s$ with $\tilde{\chi} = \chi|_U \circ \rho$.

Conversely, we can construct a inverse map by $(c, \chi_s) \mapsto (c \circ \rho) \cdot \chi_s$, which proves the map is an isomorphism. $\square$

Remark. The isomorphism is also topological, the retraction ρ induces a map

$$\rho^* : \mathrm{Hom}_{\mathrm{cont}}(U, \mathbb{C}^\times) \to \mathrm{Hom}_{\mathrm{cont}}(F^\times, \mathbb{C}^\times), \quad \chi \mapsto \chi \circ \rho$$

such a map is continous since local field is locally compact and Hausdorff, in this case the map defined in the lemma is a homeomorphism.

Lemma 1.7. Let $\chi : F^\times \to \mathbb{C}^\times$ be a quasi-character of a non-archimedean local field F, then there exists a integer $n \geq 0$ such that χ is trivial on $1 + \mathfrak{p}^n$.

Proof. $\mathbb{S}^1$ is a group has NSS (no small subgroup) properties, then there exists a neighborhood V of 1 in $\mathbb{S}^1$ such that V does not contain any non-trivial subgroup, then $\chi^{-1}(V)$ is a open neighborhood of 1 in $F^\times$. In fact, all open subgroups of $\mathcal{O}^\times$ form a chain as following:

$$\mathcal{O}^\times \supsetneq 1 + \mathfrak{p} \supsetneq 1 + \mathfrak{p}^2 \supsetneq \cdots$$

and form a basis of open neighborhood of 1 in $F^\times$, so there exists a integer $n \geq 0$ such that $1 + \mathfrak{p}^n \subset \chi^{-1}(V)$, which implies that χ is trivial on $1 + \mathfrak{p}^n$. $\square$

Hence there is no confusion to take the samllest integer $\mathfrak{f}$ and define it as the **conductor** of quasi-character χ, with a convention that $1 + \mathfrak{p}^0 = \mathcal{O}^\times$, then χ is unramified if and only if its conductor is 0, and such a concept measures how far a quasi-character is away from being unramified.

Remark. Let $F = \mathbb{Q}_p$ and $\chi : F^\times \to \mathbb{C}^\times$ be ramified character, and then $\chi|_{\mathbb{Z}_p^\times}$ factors through $\mathbb{Z}_p^\times/(1+p^{\mathfrak{f}}\mathbb{Z}_p)$ with conductor $\mathfrak{f} > 0$ as following

$$\mathbb{Z}_p^\times \longrightarrow \mathbb{Z}_p^\times/(1+p^{\mathfrak{f}}\mathbb{Z}_p) \xrightarrow{\sim} (\mathbb{Z}/p^n)^\times \xrightarrow{\varphi} \mathbb{S}^1$$

the proof is omitted here, and the interesting result is that the ramified part of χ is related to a primitive dirichlet character φ modulo $p^{\mathfrak{f}}$.

Later we will see that the local zeta integral will be defined on the character group, so it is better to treat it as a complex manifold such that we can talk about the meromorphic continuation and functional equation.

Definition 1.8. Let χ and χ' be two quasi-character of $F^\times$, they are called equivalent if $\chi'|_U = \chi'|_U$, or equivalently, χ/χ' is unramified.

It defines a equivalence relation on $\mathcal{X}(F^\times)$, and the equivalence class of a quasi-character is $[\chi] = \{\chi|\cdot|^s : s \in \mathbb{C}\} = \chi \cdot \mathcal{U}(F^\times)$, and lemma 1.5 shows that $\mathcal{U}(F^\times)$ is indeed a complex lie group of dimension 1, so each class can be identified as a Riemann surface by translation of topological group, and thus $\mathcal{X}(F^\times)$ can be identified as a disjoint union of Riemann surfaces.

Finally, to consider the continuation of local zeta function, Tate introduces a notation to denote the "dual" of a quasi-character as following:

Definition 1.9. Let $\chi \in \mathcal{X}(F^\times)$, its **twisted dual** is defined as $\chi^\vee = \chi^{-1}|\cdot|$.

It is not hard to check that $\chi \mapsto \chi^\vee$ is a homeomorphism on $\mathcal{X}(F^\times)$ with $(\chi^\vee)^\vee = \chi$, and the image of a connected component $[\chi]$ is exactly the component $[\chi^{-1}]$.

1.3 Local Zeta Function

Now we will consider the integrand on $F^\times$, which will be used to define the local zeta integral.

Lemma 1.8. Let dx be the self-dual Haar measure on F defined in the previous, then

(a) $m(E) = \int_E \frac{dx}{|x|}$ defines a Haar measure on multiplicative group $F^\times$, and when F is non-archimedean, $\mathcal{O}^\times$ gets measure $(1 - N(\mathfrak{p})^{-1})N(\mathfrak{d})^{-1/2}$.

(b) A function f is integrable on $F^\times$ with respect to $d^\times x$ if and only if $x \mapsto f(x)/|x|$ is integrable on $F - 0$ with respect to dx.

Proof. □

Example 1.9. If the number field is $\mathbb{Q}$, then by Ostrowski's theorem, the local field is either $\mathbb{R}$ or $\mathbb{Q}_p$ with p a prime number, so there is no confusion to define an integrand for a proper function f and a complex number s by

$$Z(f,s) = \int_{F^\times} f(x)|x|^s \, d^\times x$$

If $F = \mathbb{Q}$ and we choose $f(x) = e^{-\pi x^2}$, then we can calculate

$$Z(f,s) = 2\int_0^\infty e^{\pi x^2} x^{s-1} \, dx = \pi^{-s/2}\Gamma(s/2)$$

If $F = \mathbb{Q}_p$ and we choose $f(x) = \mathbf{1}_{\mathbb{Z}_p}$, then use the fact that $\mathbb{Z}_p - \{0\} = \bigcup_{n\geq 0} p^n\mathbb{Z}_p^\times$ we can calculate

$$\begin{aligned} Z(f,s) &= \sum_{n\geq 0} \int_{p^n\mathbb{Z}_p^\times} |x|_p^s \, d^\times x \\ &= \sum_{n\geq 0} p^{-ns} \int_{\mathbb{Z}_p^\times} d^\times x \\ &= \sum_{n\geq 0} p^{-ns} = \frac{1}{1-p^{-s}} \end{aligned}$$

Pay attention that in both case, the last equality holds when $\mathrm{Re}(s) > 0$, and by Euler's product formula we can find that the proudct of all local zeta functions is so-called completed zeta function

$$s \mapsto \pi^{-s/2}\Gamma(s/2)\zeta(s)$$

which plays an important role in the functional equation of Riemann zeta function, it can go back to Riemann's famous short paper [1].

In the example above, variable s can be seen as a parameter of unramified quasi-character $\chi_s = |\cdot|^s$, and beyond the example, we should consider other type of zeta function. For instance, a dirichlet L-function asspciated with a dirichlet character χ, which is defined by the following Euler product

$$L(s,\chi) = \prod_{p \text{ prime}} \frac{1}{1-\chi(p)p^{-s}}$$

it suggests that we should consider all type of quasi-character.

Definition 1.10. Fix a function $f \in \mathcal{S}(F)$, the local zeta function associated with f is defined as a integrand on space $\mathcal{X}(F^\times)$ by

$$Z(f,\chi) = \int_{F^\times} f(x)\chi(x)\, d^\times x$$

Let χ be a quasi-character of the class $[\chi_0]$ of the form $\chi_0|\cdot|^s$, it is equivalent to write

$$Z(f,\chi_0,s) = \int_{F^\times} f(x)\chi_0(x)|x|^s d^\times x$$

and we call here s as the complex parameter of χ and $\sigma = \mathrm{Re}(s)$ as the **exponent** of χ.

Review that a function on Riemann surface $f : X \to \mathbb{C}$ is called holomorphic (resp. meomorphically) at $s \in X$ if there exists a local chart (U,ϕ) with $s \in U$ such that $f \circ \phi^{-1}$ is holomorphi (resp. meomorphically) at $\phi(s)$ as a complex function.

Lemma 1.10. Following the notation above, $\chi \mapsto Z(f,\chi)$ defines a holomorphic function on the domain of quasi-character with exponent $\sigma > 0$.

Proof. We assume that K is a compact subset of the domain, for the archimedean case, schwartz condition implies that f is bounded by a constant B and f rapidly decreasing faster than any power of $1/|x|$ at infinity, so an estimate can be done by

$$|Z(f,\chi)| \le B \int_{0<|x|\le 1} |x|^{\sigma-1}\, dx + C_N \int_{|x|\ge 1} \frac{|x|^{\sigma-1} dx}{(1+|x|)^N}$$

$\sigma > 0$ ensures the first integral is finite, and the compactness of K ensures the existence of N such that $N > \sigma$ and the second integral is finite. If F is non-archimdean, suppose that $\mathrm{supp}(f) \subset \varpi^N \mathcal{O}$ for some integer $N \ge 0$, and then

$$|Z(f,\chi)| \le C \int_{0<|x|<N(\mathfrak{p})^N} |x|^{\sigma} d^{\times}x = C \sum_{n\ge -N} \int_{\varpi^n \mathcal{O}^{\times}} |x|^{\sigma} d^{\times} x$$

and lemma 1.8 (a) allows us to set $\mathrm{Vol}(\mathcal{O}^{\times}) = (1 - N(\mathfrak{p})^{-1})N(\mathfrak{d})^{-1/2}$, then

$$|Z(f,\chi)| \le C \cdot \mathrm{Vol}(\mathcal{O}^{\times}) \sum_{n\ge -N} N(\mathfrak{p})^{-n\sigma} = C \cdot \mathrm{Vol}(\mathcal{O}^{\times}) \frac{N(\mathfrak{p})^{N\sigma}}{1-N(\mathfrak{p})^{-\sigma}}$$

the last equality holds when $\sigma > 0$ by geometric series. Finally, the inequality above is enough to show the uniform convergence of $\chi \mapsto Z(f,\chi)$ in each compact of the component with $\sigma > 0$, so in each component the function is holomorphic on the domain $\sigma > 0$. □

Lemma 1.11 (Tate)**.** Fix any two functions $f, g \in \mathcal{S}(F)$, and let χ be a quasi-character χ of exponent $0 < \sigma < 1$, then $\chi^{\vee}$ is of exponent $0 < \sigma < 1$ as well, and the following identity holds

$$Z(f,\chi)Z(\hat{g},\chi^{\vee}) = Z(\hat{f},\chi^{\vee})Z(g,\chi)$$

Proof. □

Loosely speaking, the last lemma means the quotient of $Z(f,\chi)$ and $Z(\hat{f},\chi^{\vee})$ is independent of the choice of function, so we can definie a meromorphic function on domain of quasi-character of exponent $0 < \sigma < 1$ by

$$\rho(\chi) = \frac{Z(\hat{f},\chi^{\vee})}{Z(f,\chi)}$$

the choice of f should satisfy $\chi \mapsto Z(f,\chi)$ is not zero function, and we call it ρ-factors of local field F. The analytic continuation of ρ-factors will be the key to prove the local functional equation, Tate's strategy is to calculate each case explicitly.

Example 1.12. If $F = \mathbb{R}$, by example 1.4 we have

$$\mathcal{X}(\mathbb{R}^{\times}) = [\chi_0] \sqcup [\chi_1]$$

with χ_0 the trival character and χ_1 the sign character, and they are representative of even characters and odd characters respectively. In the component $[\chi_0]$, we can choose $f_0(x) = e^{-\pi x^2}$ with $\hat{f}_0 = f_0$; In the compoenent $[\chi_1]$, the choice of f_0 is not proper since its local zeta function will vanish, so we modify it by $f_1(x) = xe^{-\pi x^2}$ with $\hat{f}_1 = i\hat{f}_1$. After some calculation we can conclude the local zeta function

$$Z(f_{\epsilon},\chi_{\epsilon,s}) = \pi^{-\frac{s+\epsilon}{2}} \Gamma\left(\frac{s+\epsilon}{2}\right)$$

and its ρ-factor

$$\rho(\chi_{\epsilon,s}) = i^{\epsilon} 2^{1-s} \pi^{s} \cos \frac{\pi(s+\epsilon)}{2} \Gamma(s)$$

for quasi-character $\chi_{\epsilon,s}(t) = \mathrm{sign}(t)^{\epsilon} |\cdot|^{s}$ of exponent $\sigma = \mathrm{Re}(s) \in (0,1)$.

Example 1.13. If $F = \mathbb{C}$, by example 1.4 we have

$$\mathcal{X}(\mathbb{C}^{\times}) = \bigsqcup_{n \in \mathbb{Z}} [c_n]$$

with $c_n(z) = (z/\|z\|)^n$ the unitary character, and we can choose schwartz functions f_n for each component $[c_n]$ by

$$f_n(z) = \begin{cases} (\bar{z})^n e^{-2\pi \|z\|^2} & n \geq 0 \\ z^{-n} e^{-2\pi \|z\|^2} & n < 0 \end{cases}$$

with fourier tranform $\hat{f}_n = i^{|n|} f_{-n}$, and in each case the local zeta function does not vanish, which allows us to calculate the local zeta function

$$Z(f_n, \chi_{n,s}) = (2\pi)^{1-s+\frac{|n|}{2}} \Gamma(s + \frac{|n|}{2})$$

and its ρ-factor

$$\rho(\chi_{n,s}) = (-i)^{|n|} (2\pi)^{1-2s} \frac{\Gamma(s + \frac{|n|}{2})}{\Gamma(1 - s + \frac{|n|}{2})}$$

for quasi-character $\chi_{n,s} = c_n |\cdot|^s$ of exponent $\sigma \in (0,1)$.

Example 1.14. If F is non-archimedean, let $\mathcal{X}_n(F^{\times})$ denote the set of quasi-characters of conductor n, then we have a decomposition

$$\mathcal{X}(F^{\times}) = \bigsqcup_{n \geq 0} \mathcal{X}_n(F^{\times})$$

when $n = 0$, $\mathcal{X}_0(F^{\times})$ is just all unramified quasi-characters $\mathcal{U}(F^{\times})$; when $n = 1$, $\mathcal{O}^{\times}/1 + \mathfrak{p}$ can be identified as the multiplicative group of resuidue field $\mathcal{O}/\mathfrak{p}$, so $\mathcal{X}_1(F^{\times})$ contains $N(\mathfrak{p}) - 1$ different classes of quasi-characters; when $n > 1$, similar argument shows that $\mathcal{X}_n(F^{\times})$ contains $N(\mathfrak{p})^{n-1}(N(\mathfrak{p})-1)^2$ different classes of quasi-characters. Hence the classes of quasi-characters are countable, and we can choose test function $f_n \in \mathcal{S}(F)$ for each class $\mathcal{X}_n(F^{\times})$ by

$$f_n(x) = \begin{cases} \psi(x) = \exp(2\pi i \{\mathrm{Tr}_{F/\mathbb{Q}_p}(x)\}) & \text{if } x \in \mathfrak{d}^{-1}\mathfrak{p}^{-n} \\ 0 & \text{otherwise} \end{cases}$$

Lemma 1.15.

$$\hat{f}_n(x) = \begin{cases} (N(\mathfrak{d})^{1/2} N(\mathfrak{p})^n) & \text{if } x \in 1 + \mathfrak{p}^n \\ 0 & \text{otherwise} \end{cases}$$

Proof. □

Corollary 1.16. The ρ-factor of the unramified quasi-character is

$$\rho(|\cdot|^s) = N(\mathfrak{d})^{s-1/2} \frac{1 - N(\mathfrak{p})^{s-1}}{1 - N(\mathfrak{p}^{-s})}$$

If c_n is a unitary character of conductor $n > 0$, the ρ-factor of the ramified quasi-character is

$$\rho(c_n|\cdot|^s) = N(\mathfrak{d}\mathfrak{p}^n)^{s-\frac{1}{2}} W(\chi)$$

with root number $W(\chi)$ defined by

$$W(\chi) = N(\mathfrak{p})^{-n/2} \sum_{\bar{a} \in \mathcal{O}^\times / 1+\mathfrak{p}^n} \chi(a)\psi(\frac{a}{\varpi^{d+n}})$$

Theorem 1.17 (Tate's Local functional equation)**.**
Let F be a local field and fix a function $f \in \mathcal{S}(F)$, then $\chi \mapsto Z(f, \chi)$ has a meromorphic continuation to whole space $\mathcal{X}(F^\times)$, and there exists a meromorphic function $\rho : \mathcal{X}(F^\times) \to \mathbb{C}$ such that for any quasi-character χ such that , the following functional equation holds

$$Z(\hat{f}, \chi^\vee) = \rho(\chi) Z(f, \chi)$$

where ρ is independent of the choice of f and

Proof. In each case of local field, $\rho(\chi) = Z(\hat{f_C}, \chi^\vee)/Z(f_C, \chi)$ with f_C choosen as above examples, so ρ has a meomorphic continuation to whole space $\mathcal{X}(F^\times)$. Let $h(\chi) = \rho(\chi)Z(f, \chi)$, lemma 1.10 shows that h is meomorphic on the domain of exponent $\sigma > 0$, and by lemma 1.11 we can calculate that $h(\chi) = Z(\hat{f}, \chi^\vee)$ in the domain of exponent $0 < \sigma < 1$, and $\chi \mapsto Z(\hat{f}, \chi^\vee)$ defines a holomorphic function on the domain of exponent $\sigma < 1$. Hence the continuation of $\chi \mapsto Z(f, \chi)$ can be concluded by the continuation of h by the uniqeness of meromorphic continuation. □

2 Global Functional Equation

2.1 Adele Ring

Firstly we introduce a construction of the topological space, which allows us to obtain a finer topology than the product topology.

Definition 2.1. Let $\{X_i\}_{i\in I}$ be a family of topological spaces, and $\{U_i\}_{i\in I}$ be a family of open subset of X_i, then the **the restricted direct product of $\{X_i\}_{i\in I}$ with respect to** $\{U_i\}_{i\in I}$ is defined as a subspace of direct product $\prod_{i\in I} X_i$ as following

$$\prod_{i\in I}{}' (X_i, U_i) := \{(x_i)_{i\in I} \in \prod_{i\in I} \mid x_i \in U_i \text{ for almost all } i \in I\}$$

it can be viewed as a topological space with the open basis given by

$$\mathcal{B} = \{\prod_{i\in I} V_i \mid V_i = U_i \text{ for almost all } i \in I\}$$

If each X_i is a measurable space with σ-alegbra $\mathcal{A}_i$ and a measure μ_i, if $U_i \in \mathcal{A}_i$ for each i and $\mu_i(U_i) = 1$ for almost all i, then we can define a measure on the restricted direct product as following: Let $\mathcal{A}$ be the σ-algebra of restricted direct product generated by subsets of the form

$$M = \prod_{i \in I} M_i \quad M_i \in \mathcal{A}_i \text{ and } M_i = U_i \text{ for almost all } i \in I$$

then we define the μ on generated sets by

$$\mu(M) := \prod_{i \in I} \mu_i(M_i)$$

then μ can be extended to a measure by the Carathéodory extension theorem, and it is called the **restricted product measure** of $\{\mu_i\}_i$ with respect to $\{U_i\}_i$.

Lemma 2.1. Let $\{G_i\}_{i \in I}$ be a family of locally compact abelian groups, and $\{U_i\}_{i \in I}$ be a family of compact-open subgroups $U_i \subset G_i$, then

(a) the restricted direct product $G = \prod'_{i \in I}(G_i, U_i)$ is a locally compact abelian group under the pointwise operation.

(b) If μ_i is a Haar measure for each G_i and $\mu_i(U_i) = 1$ for almost all i, then the restricted product measure μ is a Haar measure on G.

Proof. $\square$

with the above lemma, we can formally define the adele ring.

Definition 2.2. Let K be a number field, then the adele ring $\mathbb{A}$ of K is defined as the restricted direct product of $\{K_v\}_{v \in V(K)}$ with respect to $\{\mathcal{O}_v\}_{v \in V(K)}$, where $\mathcal{O}_v = K_v$ if v is archimedean, and $\mathcal{O}_v$ is the valuation ring of K_v if v is non-archimedean. The operation of addition and multiplication on $\mathbb{A}_K$ are defined pointwisely by

$$(ab)_v = a_v b_v, \quad (a+b)_v = a_v + b_v$$

For any $a, b \in \mathbb{A}_K$.

Remark. It is a indeed a topological ring by above definition, in particular $\mathbb{A}_K$ is a locally compact abelian group under addition, and we stress it as adele group to distinguish.

We can only consider the finite place of K, that means the restricted direct product of $\{K_v\}_{v|\infty}$ with repect to $\{\mathcal{O}_v\}_{v|\infty}$, which is denoted by $\mathbb{A}_f$ and call it **the finite adele ring** of K. Review that the minkowski space K_∞ of K is defined by the embedding of K into $\mathbb{R}^{[K:\mathbb{Q}]}$, and $K_\infty \cong K \otimes_{\mathbb{Q}} \mathbb{R} \cong \prod_{v \nmid \infty} K_v$, then the adele ring can be decomposed as the direct product

$$\mathbb{A}_K \cong K_\infty \times \mathbb{A}_{K,f} \cong K \otimes_{\mathbb{Q}} \mathbb{A}_{\mathbb{Q}}$$

There exists a natural embedding of K into $\mathbb{A}_K$ by diagonal map $a \mapsto (a_v)_v$ with $a_v = a$ for any place v, it is a continous ring homomorphism, so it allows us to see K as an additive subgroup of $\mathbb{A}_K$; restrict the diagonal map on $K^\times$, it allows to see $K^\times$ as a multiplicative subgroup of $\mathbb{A}_K^\times$.

Lemma 2.2. K and $K^\times$ are discrete subgroups of $\mathbb{A}_K$ and $\mathbb{A}_K^\times$ respectively.

Proof. $\square$

Theorem 2.3 (Weak Approximation Theorem)**.**

Proof. ∎

Corollary 2.4. The quoitent groups $\mathbb{A}_K/K$ is compact

Proof. ∎

From now on, for any local field K_v, we fix dx_v to be the self-dual Haar measure in the context of lemma 1.3, notice that $K_v/\mathbb{Q}_{p_v}$ is unramified for almost all finite place v, so $\mathcal{O}_v$ gets measure $N(\mathfrak{d}_v)^{-1/2} = N(\mathcal{O}_v)^{-1/2} = 1$ for almost all finite places v, then lemma 2.1 implies a Haar measure on $\mathbb{A}$, and we denote it by dx.

Definition 2.3. Let $x = (x_v)_v \in \mathbb{A}_K$, then the adele norm of x is defined as

$$\|x\|_{\mathbb{A}_K} = \prod_v |x_v|_v$$

where $|\cdot|_v$ is the module (rf. lemma 1.1) on local field K_v

Remark. Let S be the set of places v such that $x_v \notin \mathcal{O}_v$, then S must be finite cotaining all archimedean places, and

$$\|x\|_{\mathbb{A}_K} = \prod_{v\in S} |x_v|_v \cdot \prod_{v\notin S} |x_v|_v \leq \prod_{v\in S} |x_v|_v < \infty$$

which measn the adele norm is well-defined. Morover, the norm of a non-zero element x can be zero, so it is not a usual norm.

the adele norm is multiplicative, so it induces a homomorphism on the idele group

$$\mathbb{A}_K^\times \to \mathbb{R}_{>0}, \quad x \mapsto \|x\|_{\mathbb{A}_K}$$

and its kernel is denoted by $\mathbb{A}_1^\times$, which allows us to conclude some importnat properties of the idele group.

Proposition 2.5. Along the notation above

(a) $\int_{aE} dx = |a|_{\mathbb{A}} \int_E dx$

(b) (Artin's product formula) For any $x \in K^\times$, $\|x\|_{\mathbb{A}_K} = 1$.

(c) $A_1^\times/K$ is compact.

(d) The following sequence splits

$$1 \to \mathbb{A}_1^\times/K^\times \to \mathbb{A}_K^\times/K^\times \xrightarrow{\|\cdot\|_{\mathbb{A}_K}} \mathbb{R}_{>0} \to 1$$

Proof. ∎

2.2 The Adelic harmonic analysis

This section is analogous to the section 1.2, that means the fourier analysis will be developed on the adele ring, so firstly the dual group structure should be determined.

Definition 2.4. Let $\{f_v\}_v$ be a family of functions $f_v : K_v \to \mathbb{C}^\times$ such that $f_v|_{\mathcal{O}_v} = 1$ for all but finitely many v, then for any $x = (x_v)_v \in \mathbb{A}_K$, a product of $\{f_v\}_v$ is defined as following

$$(\prod_v f_v)(x) := \prod_v f_v(x_v)$$

The condition on the family of functions is necessary to ensure the product is finite, so the product defines a function on $\mathbb{A}_K$.

Definition 2.5. A Schwartz-Bruhat function on $\mathbb{A}_K$ is defined as the product $\prod_v f_v$, with each $f_v \in \mathcal{S}(K_v)$ and $f_v = \mathbf{1}_{\mathcal{O}_v}$ for almost all finite place v. And the space of Schwartz-Bruhat functions on $\mathbb{A}_K$ is denoted by $\mathcal{S}(\mathbb{A}_K)$.

From now on, we fixe ψ_v to be the standard character on K_v in the context of definition 1.6, and $\psi = \prod_v \psi_v$. As we have seen in the section 1.2, each local field K_v is self-dual, and the dual group of $K_v/\mathcal{O}_v$ is $\mathcal{O}_v$ if v is non-archimedean, then an similar consequence is that $\mathbb{A}_K$ is self-dual.

Theorem 2.6. The map

$$\Psi : \mathbb{A}_K \to \widehat{\mathbb{A}_K}, \quad a \mapsto \psi(a-) = \prod_v \psi_v(a_v-)$$

gives an isomorphism of LCA groups.

Proof. $\square$

Corollary 2.7. The dual group of K is isomorphic to $\mathbb{A}_K/K$.

Proof. $\square$

The analogue of $\mathbb{Z} \subset \mathbb{R}$ and $K \subset \mathbb{A}_K$ allows us to develop the adelic harmonic analysis.

Definition 2.6. Let $f \in \mathcal{S}(\mathbb{A}_K)$, the fourier transform of f is defined as integrand

$$\hat{f}(y) := \int_{\mathbb{A}_K} f(x)\psi(xy)\, dx$$

where $dx = \prod_v dx_v$ is the product of self-dual Haar measure in the context of proposition 1.3.

Notice that $K_v/\mathbb{Q}_{p_v}$ is unramified for almost all finite place v, so $\mathcal{O}_v$ gets measure $N(\mathfrak{d}_v)^{-1/2} = N(\mathcal{O}_v)^{-1/2} = 1$ for almost all finite places v, hence dx here is exactly a Haar measure by lemma 2.1. In fact, the measure dx choosen here is self-dual, and it is enough to choose a nice function $f \in S(\mathbb{A}_K)$ to verify.

Theorem 2.8 (Poisson summation formula)**.** If $f \in \mathcal{S}(\mathbb{A})$, then

$$\sum_{a\in K} f(a) = \sum_{a\in K} \hat{f}(a)$$

2.3 Idele Group and Hecke Character

The unit group of the adele ring $\mathbb{A}_K^\times$ is not a topological group (the subspace topology of $\mathbb{A}_K$) under the multiplication, the reason is that the inverse map $(a_v)_v \mapsto (a_v^{-1})_v$ is not continous, so a finer topology should be given to the unit group for a further study.

Definition 2.7. The idele group of K is defined as the restricted direct product of $K_v^\times$ with respect to $\mathcal{O}_v^\times$, where $\mathcal{O}_v^\times = K_v^\times$ if v is archimedean, and $\mathcal{O}_v^\times$ is the unit group of $\mathcal{O}_v$ if v is non-archimedean.

Remark. It is a multiplicative locally compact abelian group, and analogously to the adele ring, **the finite idele group** of K is defined as the restricted direct product of $\{K_v^\times\}_{v\nmid\infty}$ with respect to $\{\mathcal{O}_v^\times\}_{v\nmid\infty}$ for all non-archimedean place v, which is denoted by $\mathbb{A}_{K,f}^\times$, and idele group can be decomposed as the direct product

$$\mathbb{A}_K^\times \cong K_\infty^\times \times A_{K,f}^\times$$

As local multiplicative characters defined in the section 1.3, we shall also develop the global multiplicative characters such that we can define the global zeta function.

Lemma 2.9. Let $\chi : \mathbb{A}^\times \to \mathbb{C}^\times$ be a continous homomorphism, define the local quasi-character at each place v by

$$\chi_v : K_v \to \mathbb{C}^\times, \quad a_v \mapsto \chi(1, \ldots, \underbrace{a_v}_{v \text{ place}}, \ldots, 1)$$

then $\chi = \prod_v \chi_v$ and there eixists a finite subset S of places, it contains all archimdean places and χ_v is unramified for any $v \notin S$.

Proof. The product $\prod_v \chi_v$ is always finite by the definition of idele group, and an algebraic calculation shows it is equal to χ. Let $\mathbb{A}_f^\times$ be the finite idele group, and then consider the restriction of χ on $\mathbb{A}_f^\times$, and denote it by χ_f. Finite idele group is a totally disconnected LCA group, so by NSS property of $\mathbb{C}^\times$, we can conclude a open neighborhood U of identity in $\mathbb{A}_f^\times$ such that χ_f is trival on U, and then the topology of $\mathbb{A}_f^\times$ implies the existence of a finite subset S of places such that

$$\prod_{v\in S} U_v \times \prod_{v\notin S} \mathcal{O}_v^\times \subset U, \quad v \text{ finite}$$

where U_v is an open subgroup of $K_v^\times$. Adding all archimedean places into S, we can conclude that χ_v is unramified for any $v \notin S$. $\square$

Definition 2.8. A **Hecke character** or a **quasi-character** of K is a continous homomorphism $\chi : \mathbb{A}_K^\times \to \mathbb{C}^\times$ such that $\chi|_{K^\times} = 1$, and its twisted dual is defined by $\chi^\vee = \chi^{-1}|\cdot|_{\mathbb{A}}$.

Remark. The idele class group of K is defined as quotient group $C_K = \mathbb{A}^\times / K^\times$, so equivalently, a Hecke character is a continous homomorphism on the idele class group.

By product formula in proposition 2.5, we can see that $|\cdot|_{\mathbb{A}}^s$ defines a Hecke character for any $s \in \mathbb{C}$, and there are a condition to determine whether a Hecke character is of this form, which is analogous to lemma 1.5 to determine unramified local quasi-character.

Lemma 2.10. A Hecke character $\chi : \mathbb{A}^\times \to \mathbb{S}^1$ is of the form $|\cdot|_{\mathbb{A}}^s$ for some $s \in \mathbb{C}$ if and only if χ is trivial on subgroup $\mathbb{A}_1^\times$.

Proof. $\mathbb{A}_1^\times$ is the kernel of $|\cdot|_{\mathbb{A}}$, so it is enough to prove the necessity. If $\mathbb{A}_1^\times \subset \ker\chi$, then χ factors through the quotient group $\mathbb{A}^\times/\mathbb{A}_1^\times$ such that a diagram commutes

$$\begin{tikzcd} \mathbb{A}^\times \arrow[r, "\chi"] \arrow[d] & \mathbb{C}^\times \\ \mathbb{R}_{>0} \cong \mathbb{A}^\times/\mathbb{A}_1^\times \arrow[ur, dotted, "\bar{\chi}"'] & \end{tikzcd}$$

then the result is immediate by the quasi-character of $\mathbb{R}_{>0}$. ☐

Lemma 2.11. For any Hecke character χ, there exists a complex number s and a unitary Hecke character η such that $\chi = \eta|\cdot|_{\mathbb{A}}^s$.

Proof. ☐

Similarly, two Hecke chracters χ and χ' on class group $A^\times/K$ are called equivalent if they concide on the compact subgroup $A_1^\times/K$, then each equivalence class consists of Hecke characters of the form $\eta|\cdot|_{\mathbb{A}}^s$ with η a fixed unitary character, and a parametrization can be done to identify each class with complex plane, which shows that the set of Hecke characters has the structure of Riemann surface with countable connected components homeomorphic to $\mathbb{C}$, hence $\chi \mapsto \chi^\vee$ gives an biholomorphic involution on the space of Hecke characters.

2.4 Global Functional Equation

Definition 2.9. Let $f \in \mathcal{S}(\mathbb{A})$ and χ be a Hecke character, then the global zeta function associated to f and χ is defined by

$$Z(f, \chi) := \int_{\mathbb{A}^\times} f(x)\chi(x)\, d^\times x$$

We should restrict the domain to be one equivalence class of hecke characters, that means if we take a class $[\chi_0]$ with χ_0 a unitary Hecke character, then a parametrization should be

$$Z(f, \chi) = Z(f, \chi_0, s) = \int_{\mathbb{A}} f(x)\chi_0(x)|x|_{\mathbb{A}}^s\, dx$$

that allows us to view it as a integrand on complex plane.

Lemma 2.12. $\chi \mapsto Z(f, \chi)$ defines a holomorphic function on the domain of exponent $\sigma > 1$.

Proof. ☐

Claim 2.13 (Quotient integral formula)**.** Let G be a LCA group and H its closed subgroup, then for any two Haar measure μ_G and μ_H on G and H respectively, there exists a unique Haar measure $\mu_{G/H}$ on the quotient group G/H such that for any $f \in L^1(G)$, we have

$$\int_G f(g)d\mu_G(g) = \int_{G/H} \left(\int_H f(gh)d\mu_H(h) \right) d\mu_{G/H}(gH)$$

A little work should be done to rewrite the global zeta function such that we can imitate the Riemann's proof of zeta function. Notice that $\mathbb{A}^\times/\mathbb{A}_1^\times \cong \mathbb{R}_{>0}$ as locally compact abelian groups, so by quotient inetral formula there exits a Haar measure d^*x on closed subgroup $\mathbb{A}_1^\times$ such that $d^\times x = d^*x \cdot dt/t$, so we can see the each $t > 0$ as an idele element by

$$\tilde{t} = (\underbrace{t^{1/N}, \ldots, t^{1/N}}_{\text{archimdean place}}, 1, ...), \quad N = [K : \mathbb{Q}]$$

and then global zeta function can be rewritten as following

$$Z(f, \chi) = \int_0^\infty Z_t(f, \chi) \frac{dt}{t} \tag{1}$$

with auxiliary function defined by

$$Z_t(f, \chi) = \int_{\mathbb{A}_1^\times} f(\tilde{t}x)\, \chi(\tilde{t}x) d^*x$$

Hence now we can consider the symmerty of auxiliary function, it will be the key to finish the meomorphic continuation.

Lemma 2.14. $\chi \mapsto Z_t(f, \chi)$ defines a holomorphic function on the domain of Hecke characters, and for any $t > 0$ we have

$$Z_t(f, |\cdot|^s) = Z_{1/t}(\hat{f}, |\cdot|^{1-s}) + \frac{\hat{f}(0)V}{t^{1-s}} - f(0)Vt^s$$

where V is the volume of fundamental domain of $\mathbb{A}_1^\times/K^\times$ with respect to the measure d^*x, and for any Hecke character χ not equivalent to $|\cdot|^s$, we have

$$Z_t(f, \chi) = Z_{1/t}(\hat{f}, \chi^\vee)$$

Proof. Firstly we calculate the general equation of $Z_t(f, \chi)$, $K^\times$ can be identified as a discrete subgroup of $\mathbb{A}^\times$, so there exists a Haar measure $d^\star x$ on $\mathbb{A}_1^\times/K^\times$, such that

$$Z_t(f, \chi) = \int_{\mathbb{A}_1^\times/K^\times} \sum_{a \in K^\times} f(a\tilde{t}x)\chi(a\tilde{t}x) d^\star x$$

Hecke character is trival on $K^\times$, so $\chi(a\tilde{t}x) = \chi(\tilde{t}x)$ is independent of a, and we add the term $a = 0$ such that we can apply the adelic Poisson summation formula

$$\begin{aligned} Z_t(f, \chi) + f(0) \int_{\mathbb{A}_1^\times/K^\times} \chi(\tilde{t}x) d^\star x &= \int_{\mathbb{A}_1^\times/K^\times} \sum_{a \in K} f(a\tilde{t}x)\chi(\tilde{t}x) d^\star x \\ &= \int_{\mathbb{A}_1^\times/K^\times} \frac{1}{t|x|_{\mathbb{A}}} \sum_{a \in K} \hat{f}(a/\tilde{t}x)\chi(\tilde{t}x) d^\star x \end{aligned}$$

a change variable $x = y^{-1}$ allows us to calcluate the functional equation for any $t > 0$

$$Z_t(f, \chi) + f(0) \underbrace{\int_{\mathbb{A}_1^\times/K^\times} \chi(\tilde{t}x) d^\star x}_{I(\chi)} = Z_{1/t}(\hat{f}, \chi^\vee) + \hat{f}(0) \int_{\mathbb{A}_1^\times/K^\times} \chi^\vee(x/\tilde{t}) d^\star x \tag{2}$$

hence it is enough to discuss the integration

$$I(\chi) = \chi(\tilde{t}) \int_{\mathbb{A}_1^\times / K^\times} \chi(x) d^\star x$$

If $\chi = |\cdot|_{\mathbb{A}}^s$, then $I(\chi) = t^s V$ clearly; Otherwise, χ is not trival on $\mathbb{A}_1^\times / K^\times$ by lemma 2.10, and then compactness of $\mathbb{A}_1^\times / K^\times$ implies $I(\chi) = 0$. Finally, we plug the result into the equation 2 to finish the proof. □

Theorem 2.15 (Hecke-Tate)**.** Fix $f \in \mathcal{S}(\mathbb{A})$, then $\chi \mapsto Z(f, \chi)$ has a meomorphic continuation to the whole domain of Hecke characters, and it has only two simple poles at $|\cdot|^0$ and $|\cdot|^1$ with residue $-f(0)V$ and $\hat{f}(0)V$ respectively. Finally, it satisfies the functional equation for any other charecters

$$Z(f, \chi) = Z(\hat{f}, \chi^\vee)$$

Proof. We separate the integral (equation 2.4) into two parts

$$Z(f, \chi) = \int_0^1 Z_t(f, \chi) \frac{dt}{t} + \int_1^\infty Z_t(f, \chi) \frac{dt}{t} \tag{3}$$

the second part is holomorphic on each component of character space since f decreases rapidly at infinity, so it is enough to evaluate the first part by changing variable $t \mapsto 1/t$, and applying the lemma 2.14 to arrange the integral. If $\chi = |\cdot|_{\mathbb{A}}^s$, then we can get

$$Z(f, |\cdot|_{\mathbb{A}}^s) = \frac{\hat{f}(0) \cdot V}{s-1} - \frac{f(0) \cdot V}{s} + \int_1^\infty [Z_t(f, \chi) + Z_t(\hat{f}, \chi^\vee)] \frac{dt}{t}$$

which gives the meomiorphic continuation of $\chi \mapsto Z(f, \chi)$ on the component of $|\cdot|_{\mathbb{A}}$, and the pole is given by first two terms; If χ is not of the form, similarly we can get

$$Z(f, \chi) = \int_1^\infty [Z_t(f, \chi) + Z_t(\hat{f}, \chi^\vee)] \frac{dt}{t}$$

which gives the holomorphic continuation on the other components. Finally, in both cases, the functional equation holds after a simple calculation. □

Remark. A comparison can be done with the Riemann zeta function, the completed zeta function is defined by $\xi(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$, and a integral representation of ξ is given by

$$\xi(s) = \int_0^\infty \psi(t) t^{s/2} \frac{dt}{t}, \quad \mathrm{Re}(s) > 1$$

with

$$\psi(t) = \frac{1}{2}(\sum_{n \in \mathbb{Z}} e^{-\pi n^2 t} - 1)$$

and we can apply the poisson summation formula to get the symmetry of ψ, and then get the functional equation of ξ by diving the integral into two parts (see a proof in theorem 2.3 of chapter 6 in [**?**]). The Hecke-Tate theorem is a generalization of Riemann's proof, the role of auxiliary function $t \mapsto Z_t(f, \chi)$ is analogous to the role of ψ.

In complex analysis, the mellin transform of a function $f : \mathbb{R}_{>0} \to \mathbb{C}$ is defined by

$$\mathcal{M}[f](s) = \int_0^\infty f(t) t^s \,\frac{dt}{t}$$

the asymptotic behavior of f at 0 and ∞ determines a "analytical strip", where $s \mapsto \mathcal{M}[f](s)$ defines a holomorphic function, hence ...

If $\chi = \eta|\cdot|_{\mathbb{A}}^s$, then in equation (2.4) we notice that $\chi(\tilde{t}x) = \eta(x)t^s$, so a separtion of variable allows us to see the global zeta function on component $[\eta]$ as a mellin transform of a function on t as following

$$\Phi(t) = \int_{\mathbb{A}_1^\times} f(\tilde{t}x)\eta(x)\, d^*x$$

3 Conclusion and complement

References

[1] B. Riemann, "Ueber die anzahl der primzahlen unter einer gegebenen grosse," *Ges. Math. Werke und Wissenschaftlicher Nachlaß*, vol. 2, no. 145-155, p. 2, 1859.